**1. Primary Alignment: Items 2, 6, and 4**

- **Items 2 & 6: Identify the key variables in a model / Identify and describe the mathematical aspects of a real-world problem... identifying the significant variables**
  - **Analysis:** This is the foundational step. Students must extract all critical variables from the text: rotation speed (4 rpm), number of sectors (3), capacity per sector (2 people), and total time (30 minutes) (p. 1). Identifying these variables and understanding how they interact is the prerequisite for building the mathematical model.
- **Item 4: Read, decode and make sense of statements, questions, tasks, objects or images to create a model of the situation**
  - **Analysis:** Students must decode the written information and combine it with the diagrams to understand the physical process of "entering the building" (p. 1). They must realize that "one rotation" means "three sectors" pass the entrance, thereby converting physical movement into a calculable mathematical model.

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**2. Significant Alignment: Items 9 and 5**

- **Item 9: Translate a problem into a standard mathematical representation or algorithm**
  - **Analysis:** After identifying the variables, students translate the logic into a standard arithmetic algorithm: $4\ (\text{rotations/min}) \times 3\ (\text{sectors/rotation}) \times 2\ (\text{people/sector}) \times 30\ (\text{minutes})$. This demonstrates the ability to translate a real-world scenario into a continuous multiplication algorithm.
- **Item 5: Recognise mathematical structure (including regularities, relationships, and patterns) in problems or situations**
  - **Analysis:** The problem reflects a classic "rate times time" mathematical structure. Students need to recognize the linear relationship between the compound rate (people per minute) and the total time elapsed.

**3. Key Logical Alignment: Items 7, 12, and 8**

- **Item 7: Simplify or decompose a situation or problem in order to make it amenable to mathematical analysis**
  - **Analysis:** Students decompose the 30-minute problem. A typical approach is to calculate people per rotation, then people per minute, and finally the total for 30 minutes. This decomposition makes the complex real-world problem easy to analyze.
- **Item 12: Identify the constraints, assumptions simplifications in a mathematical model**
  - **Analysis:** The question asks for the "**maximum**" number of people (p. 1). This implies a modeling assumption: every sector in every rotation for the full 30 minutes must be at full capacity (2 people). Identifying this constraint is key to reaching the correct answer (D. 720).
- **Item 8: Recognise aspects of a problem that correspond with known problems or mathematical concepts, facts or procedures**
  - **Analysis:** Students must recognize that this is essentially a multi-level unit conversion or rate problem, similar to production or flow-rate problems they have encountered in math class.

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**4. Secondary Alignment: Items 10, 1, 3, and 11**

- **Analysis:** These items have lower relevance. Since the diagrams and variables are already provided (p. 1), students are primarily utilizing existing information rather than "selecting" or "creating" a new mathematical representation (Items 1, 3) to describe the problem from scratch.

**1. 核心关联：第 2、6 项 与 第 4 项**

- **第 2 项 & 第 6 项：Identify the key variables in a model / Identify and describe the mathematical aspects of a real-world problem... identifying the significant variables** (p. 1)
  - **分析：** 这是解题的首要步骤。学生必须从文字描述中提取出所有关键变量：旋转速度（4 圈/分）、扇区数量（3 个）、单个扇区容量（2 人）以及总时间（30 分钟）。识别这些变量并理解它们如何相互作用是建立数学模型的前提。
- **第 4 项：Read, decode and make sense of statements, questions, tasks, objects or images to create a model of the situation** (p. 1)
  - **分析：** 学生需要解码文字信息并结合示意图来理解“进入大楼”的物理过程。他们必须理解“每一圈”意味着“三个扇区”都会经过入口，从而将物理运动转化为可计算的数学模型。

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**2. 重要关联：第 9 项 与 第 5 项**

- **第 9 项：Translate a problem into a standard mathematical representation or algorithm** (p. 1)
  - **分析：** 在识别变量后，学生需要将逻辑转化为标准的算术算法。即：$4(\text{圈/分}) \times 3(\text{扇区/圈}) \times 2(\text{人/扇区}) \times 30(\text{分钟})$。这体现了将现实情境翻译为连乘算法的能力。
- **第 5 项：Recognise mathematical structure (including regularities, relationships, and patterns) in problems or situations** (p. 1)
  - **分析：** 题目反映了一种典型的“速率乘以时间”的数学结构。学生需要识别出这种复合速率（人/分钟）与总时间之间的线性关系。

**3. 关键逻辑关联：第 7、12 项 与 第 8 项**

- **第 7 项：Simplify or decompose a situation or problem in order to make it amenable to mathematical analysis** (p. 1)
  - **分析：** 学生需要将 30 分钟的大问题分解。通常的思路是先算出一圈进多少人，再算出一分钟进多少人，最后得出 30 分钟的总数。这种分解过程使复杂的现实问题变得易于分析。
- **第 12 项：Identify the constraints, assumptions simplifications in a mathematical model** (p. 1)
  - **分析：** 题目问的是“最大（maximum）人数”。这隐含了一个建模假设：在 30 分钟内的每一分钟、每一圈、每一个扇区都必须是满载（2人）的状态。识别这一约束条件是得出正确答案 D (720) 的关键。
- **第 8 项：Recognise aspects of a problem that correspond with known problems or mathematical concepts, facts or procedures** (p. 1)
  - **分析：** 学生需识别出这本质上是一个多级单位换算或连乘问题，类似于他们在数学课上学过的产量计算或流速问题。

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**4. 次要关联：第 10、1、3、11 项**

- **分析：** 这些项的相关性较低。因为题目已经提供了图示和变量（第 10、11 项），学生主要是利用既定信息进行计算，而不是自己去“选择”或“创建”一个新的数学表征（第 1、3 项）来描述问题。